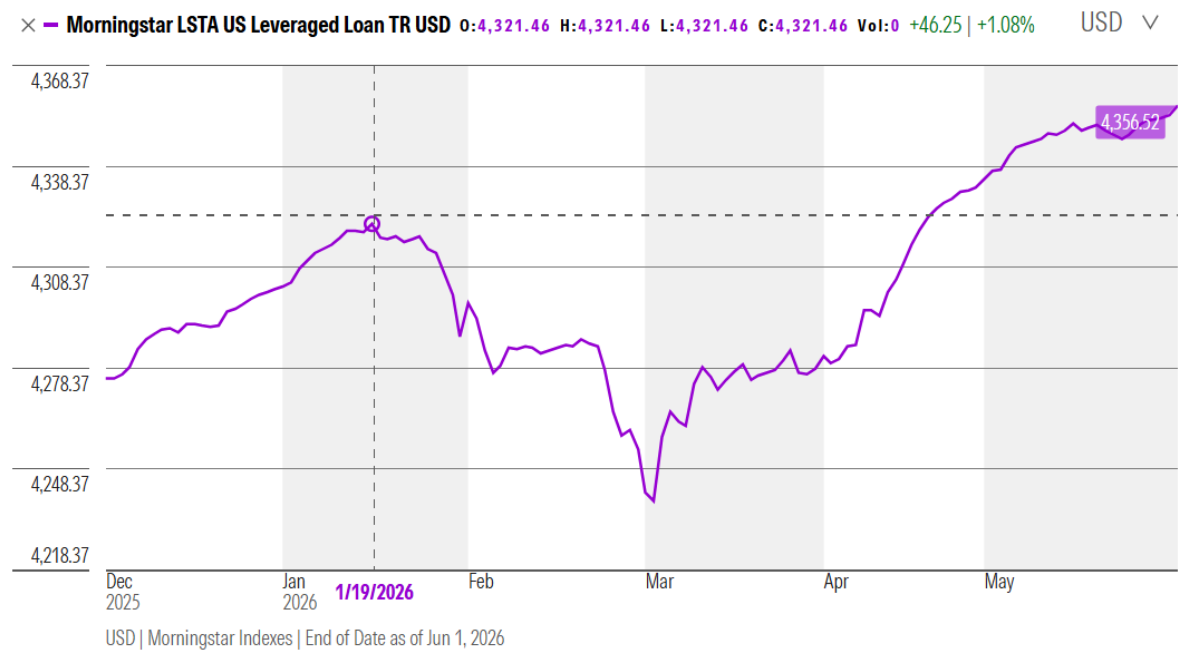


Project Description

1. Summary

This is a thesis-first LLM based pipeline for a leveraged loan portfolio manager. It fetches public market commentary, per-sector credit news, geopolitical context, and government macro data; synthesises them into a structured directional WorldView on the leveraged loan market; validates that WorldView against its own cited sources using a second LLM call; and then maps the thesis onto a real ETF portfolio to produce a ranked impact table per holding.

Leveraged loans had negative quarterly returns in Q1 2026 owing to the selloff in the software sector. BKLN (Invesco Senior Loan ETF) is an ETF that tracks the Morningstar LSTA US Leveraged Loan 100 index.



The question the project tries to answer is: given only information available before 15-Jan-2026, would the agent have identified that a portfolio that tracks the top 20 holdings of the BKLN portfolio was dangerously concentrated in the sector about to fall 700bps (Software)?

PS: The current version of the project is slow and takes about 6-7 min to fully run.

2. Methodology

The portfolio is the Invesco Senior Loan ETF (BKLN), top-20 holdings from SEC N-PORT filing accession **0001378872-26-000362**, dated **30-NOV-2025**. These 20 positions are loaded from a CSV and carry CUSIP, sector tag, floating coupon rate, and portfolio weight. Position 1 is Invesco Premier money market (~21.5% weight, excluded from credit analysis). The remaining 19 are leveraged loans across seven sectors. Software names account for nine of the nineteen and roughly 37% of the top-20 portfolio weight.

The **analysis cutoff is 15-Jan-2026**. The Morningstar/S&P LSTA LL100 index peaked four days later on 19-Jan-2026. Every source document is rejected if dated after that date — two independent date gates enforce this.

The **known outcome**, hardcoded in ``eval/ground_truth.py``:

```
GROUND_TRUTH = {  
    "overall_loan_market_bps": -184,  
    "software_loans_bps": -700,  
    "hy_spread_change_bps": +52,  
}  
  
SECTOR_OUTCOMES_BPS = {  
    "Software": -700, "Healthcare": -120, "Business Services": -150,  
    "Industrials": -90, "Consumer & Retail": -180,  
    "Financial Services": -80, "Media & Entertainment": -130,  
}
```

These numbers are sourced and estimated from pitchbook reports on the LL market.

3. The end-to-end workflow

Four fetch passes → one LLM synthesis → validation feedback loop (LLM) → pure-Python portfolio mapping → impact scoring → Eval.

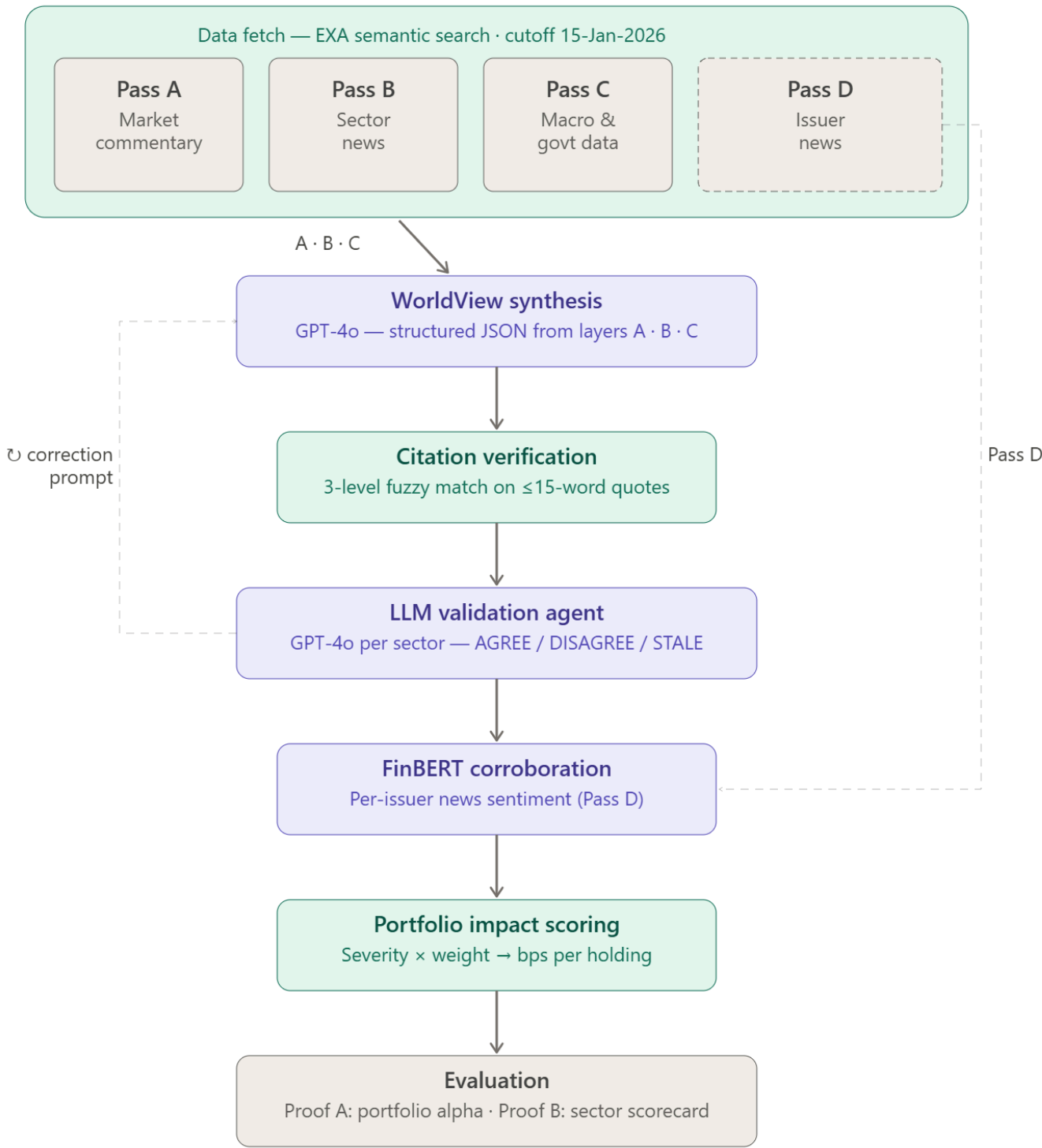
When **"Fetch Market Data"** is clicked, the server runs four fetch passes in sequence. This is done using EXA's semantic web search.

- **Pass C (macro/govt):** Lookback: 01-Dec-2025.
- **Pass A (market commentary):** Queries for credit stress signals, LSTA index data, rating actions, AI disruption news, and geopolitical risk. Lookback: 01-Jun-2025.
- **Pass B (sector news):** queries - one per WorldView sector, targeting current credit conditions in that specific sector. Lookback: 01-Dec-2025.
- **Pass D (issuer news):** One query per holding, targeting credit events for that specific borrower. Lookback: 01-Jun-2025.

"Run World View Analysis" executes one ``gpt-4o`` call that reads Pass A + B + C text concatenated, then a validation loop runs which is another LLM call to validate whether the analysis is correctly cited from the fetched documents. Source weighting rules in the prompt tell the LLM to discount asset manager outlook pieces and weight index data and rating actions highest. The LLM is told that ``[SECTOR: X]`` articles are **primary evidence** for that sector's call and outweigh general outlooks.

"Run Portfolio Mapping" - For each holding it runs FinBERT on the issuer's Pass D news and adjusts conviction.

"Run Evaluation" computes proof points against the hardcoded ground truth. This is to see if our system actually worked and produced a worthwhile result



4. Synthesis and the validation feedback loop

The prompt

SYNTHESIS_PROMPT = ""

*You are a senior credit analyst at a leveraged loan asset management firm.
Your analysis cutoff is 15-Jan-2026. Do not reference or imply knowledge of
any market events after this date.*

The text below contains FOUR distinct data layers. Read all of them:

[LAYER 1 — MACRO & RATES]

*Government sources (Fed, NY Fed, BEA): SOFR, Fed Funds rate, GDP, FOMC
decisions. These are ground truth — use them to anchor rate-sensitive calls.*

[LAYER 2 — MARKET COMMENTARY]

*LSTA/Morningstar index data, CLO secondary market, credit stress signals,
rating agency actions, leveraged loan spread and price levels. These are
quantitative anchors for overall market direction and conviction.*

[LAYER 3 — SECTOR-SPECIFIC NEWS]

*Articles tagged [SECTOR: X] contain current credit conditions, spread
movements, downgrades, and stress signals for that specific sector.
You MUST use these to make your sector call for that sector — do not
rely on general market commentary for a sector if a [SECTOR: X] article
exists.*

[LAYER 4 — GEOPOLITICAL & REGULATORY]

*Tariff risk, US-China tech policy, AI regulation, antitrust — these are
cross-sector macro risks. Incorporate them into both the overall thesis
and any sector calls where they directly apply (e.g. tariffs → Industrials,
AI/tech policy → Software, drug pricing → Healthcare).*

=== SOURCE TEXT ===

{commentary_text}

=== SOURCE WEIGHTING ===

- 1. HIGHEST: Index data (LSTA, Morningstar spreads), rating actions, secondary
market prices, government rate data. Quantitative — anchor conviction here.*
- 2. HIGH: [SECTOR: X] articles — current news about that sector's credit health.*
- 3. MEDIUM: Geopolitical/regulatory news with direct credit impact.*
- 4. LOWER: Asset manager annual outlooks (Apollo, Blackstone, KKR, Ares).
These are structurally optimistic marketing material. Only use when they
cite specific spread data or acknowledge explicit downside risks.*

=== YOUR TASK ===

Produce a structured JSON world view of the leveraged loan market as of 15-Jan-2026:

- 1. Overall directional thesis, using market-wide spread and rate data.*
- 2. Key macro factors — include geopolitical risks where they affect credit.*
- 3. Sector call for EACH of the 7 sectors — use the [SECTOR: X] articles
as PRIMARY evidence. If a sector article shows stress, that outweighs a
bullish general outlook from an asset manager.*

CONVICTION SCALE:

0.75–0.95 = high — clear quantitative evidence (spread levels, rating actions)
0.55–0.75 = moderate — directional lean from news or commentary
0.40–0.55 = low — sparse or contradictory signals; default to NEUTRAL

CITATION RULE (enforced in code — non-negotiable):

*cited_quote must be a verbatim substring of the text above, ≤15 words.
Do not paraphrase. If no good quote exists, use the closest phrase that
appears verbatim. Prefer quotes from index data or news over outlooks.*

Return a JSON object with EXACTLY this schema — all 7 sectors required:

```
{{
  "analysis_date": "2026-01-15",
  "overall_direction": "BEARISH" | "NEUTRAL" | "BULLISH",
  "overall_conviction": <float 0-1>,
  "summary": "<2-3 sentence thesis — must reference specific spread levels or credit events from the text>",
  "sector_calls": {{
    "Software": {{ "direction": "BEARISH"|"NEUTRAL"|"BULLISH", "conviction": <float>,
    "reasoning": "<1-2 sentences citing specific evidence>", "cited_quote": "<verbatim ≤15 words>" }},
    "Healthcare": {{ "direction": "BEARISH"|"NEUTRAL"|"BULLISH", "conviction": <float>,
    "reasoning": "<1-2 sentences citing specific evidence>", "cited_quote": "<verbatim ≤15 words>" }},
    "Business Services": {{ "direction": "BEARISH"|"NEUTRAL"|"BULLISH", "conviction": <float>,
    "reasoning": "<1-2 sentences citing specific evidence>", "cited_quote": "<verbatim ≤15 words>" }},
    "Industrials": {{ "direction": "BEARISH"|"NEUTRAL"|"BULLISH", "conviction": <float>,
    "reasoning": "<1-2 sentences citing specific evidence>", "cited_quote": "<verbatim ≤15 words>" }},
    "Consumer & Retail": {{ "direction": "BEARISH"|"NEUTRAL"|"BULLISH", "conviction": <float>,
    "reasoning": "<1-2 sentences citing specific evidence>", "cited_quote": "<verbatim ≤15 words>" }},
    "Financial Services": {{ "direction": "BEARISH"|"NEUTRAL"|"BULLISH", "conviction": <float>,
    "reasoning": "<1-2 sentences citing specific evidence>", "cited_quote": "<verbatim ≤15 words>" }},
    "Media & Entertainment": {{ "direction": "BEARISH"|"NEUTRAL"|"BULLISH", "conviction":
    <float>, "reasoning": "<1-2 sentences citing specific evidence>", "cited_quote": "<verbatim ≤15
    words>" }}
  }}
}}
```

*Return ONLY valid JSON. No preamble, no markdown fences, no explanation. *
""""

Citation verification

Every ``cited_quote`` (≤15 words) is verified against the source text via three-level matching:

1. Exact substring
2. Case-insensitive substring
3. Token-level fuzzy match (``SequenceMatcher ≥ 0.82``, only for quotes >4 words)

Level 3 catches minor punctuation differences that caused genuine quotes to fail exact matching. Failed quotes: ``citation_verified=False``, conviction zeroed.

5. LLM validation agent

After `\_verify\_citations()`, `\_validate\_world\_view()` runs **one `gpt-4o` call per sector** that passed R2. Each call reads ± 500 chars around the citation and answers two questions:

1. Does this paragraph **AGREE / DISAGREE / AMBIGUOUS** with the directional call?
2. Is the excerpt **primarily based on pre-2025 data** (stale)?

Penalties applied to conviction in-place:

- `DISAGREES` → `direction\_mismatch=True`, conviction -0.10
- `STALE\_DATA: YES` → conviction -0.05 (penalties stack)

The feedback loop

If any sector has `DISAGREES`, a correction prompt is built listing flagged sectors with their citation context and the validator's one-sentence reasoning. The LLM re-synthesises only those sectors. This runs at most once, giving worst-case 2 synthesis calls + 14 validation calls total.

6. FinBERT issuer corroboration

Input: first 5,400 chars of the issuer's Pass D news file. This captures the headline and opening of each article where all directional signals sit.

Processing:

1. Split text at paragraph boundaries into chunks $\leq 1,800$ chars
2. Run FinBERT on each chunk → `{positive, negative, neutral}` probabilities
3. Aggregate by character-weighted average across chunks
4. Drop neutral, renormalise positive+negative to 100%
5. If positive share $\geq 60\%$ → BULLISH; if negative $\geq 60\%$ → BEARISH; else MIXED

Mapping to corroboration:

- Same direction as sector call → `CORROBORATES`
- Opposite direction → `CONTRADICTS`
- Below 60% threshold → `MIXED`
- Sector NEUTRAL or no news file → `NO\_SIGNAL`

Conviction adjustment:

- CORROBORATES 'ISSUER_CORROBORATED' +0.10 (cap 0.95)
- CONTRADICTS 'ISSUER_CONTRADICTED' -0.15 (floor 0.0)
- MIXED 'MIXED' 0
- NO_SIGNAL (no file fetched) 'SECTOR_INFERRED' 0

7. Impact scoring (pure Python)

SEVERITY_BPS = {LOW: 25, MEDIUM: 75, HIGH: 200, CRITICAL: 400, NEUTRAL: 0}
 impact_bps = SEVERITY_BPS[severity] * sign * (weight_norm_top20 / 100)
 sign: SHORT=-1, LONG=+1, NEUTRAL=0

Severity rules:

- conviction < 0.65 → **NEUTRAL**
- SHORT + conviction ≥ 0.75 → **CRITICAL**
- conviction ≥ 0.80 → **HIGH**
- conviction ≥ 0.70 → **MEDIUM**
- conviction ≥ 0.65 → **LOW**

The CRITICAL/400bps figure is calibrated against the known software outcome (-700bps actual - sourced from Morningstar article). 400bps is ~57% of that tail — conservative enough not to claim precision on a sector-proxy basis.

8. Evaluation

Proof A — Portfolio alpha. Benchmark: weighted sector outcomes for all 20 holdings.

Agent: weighted sector outcomes for non-SHORT holdings only, with weights renormalised to 100%. Alpha = agent return minus benchmark. Positive when the exited positions underperformed.

Proof B — Sector scorecard. For each of the 7 sectors:

- BEARISH + actual < -100bps → HIT
- NEUTRAL + actual between -100 and 0bps → HIT
- BULLISH + actual > 0bps → HIT
- Otherwise MISS (or NEAR for borderline)

8. Limitations

Sector return proxies. `SECTOR\_OUTCOMES\_BPS` are sector-level estimates from PitchBook commentary, not loan-level marks. Individual loans within a sector vary significantly.

Top-20 only. BKLN holds ~186 positions. Positions 21–186 are not modelled.

Pass D fetches journalism, not financials. The issuer corroboration reads news articles, not actual financial statements, covenant compliance certificates, or Bloomberg loan marks.

Manual sector tagging. Sector labels in the CSV are manually assigned. A mislabelled issuer maps to the wrong sector call with no warning.

Validation LLM is the same model as synthesis. `gpt-4o` validating `gpt-4o` output is not fully independent. A different model family for validation would be more robust.

FinBERT text cap. Each issuer's news is capped at 5,400 chars for inference. Long articles beyond that cap are not read — a very long negative article could have material content past that point.

The evaluation for the sector hit and miss were estimated from the pitchbook commentary - a better alternative would have been historical and specifically quarterly performance of the sectors.

9. Future Scope of Improvement

Debating Agent - Instead of a source weighting embedded in the LLM call for synthesis we could have separate LLMs synthesising the 4 different data fetched and a debating agent deciding which synthesis should be weighed more in the final synthesis for a sector. For eg., if a sector level synthesis contradicts with macro level synthesis about the sector- a debating agent could help resolve this.

Multi LLM Model evaluation - The project could be run multiple times using different LLM models to compare results

RAG – The current system operates on a relatively small set of fetched documents. As data volume scales, a relevance-ranking layer can be introduced to surface the most informative

articles, while RAG can ground synthesis on retrieved evidence and further reduce hallucinations.

10. Results

While the program was able to predict the bearish sentiment around the software sector, it did so with different conviction levels upon different multiple runs which was expected because each fetch might have slightly different files and the synthesis generated by the LLM might also differ each time we run the program.

BKLN Signal Agent

PORTFOLIO: BKLN

N-PORT: 30-NOV-2025

CUTOFF: 15-JAN-2026

42 market docs fetched

Fetch Market Data

00

Exa semantic search → dated commentary files (window: 01-JUN-2025 → 15-JAN-2026)

DONE

42 docs

Analysing

38 documents accepted

42 market docs — window: 2025-06-01 → 15-JAN-2026

View Commentary

✓ <https://www.newyorkfed.org/medialibrary/media/markets/omo/omo2025-pdf.pdf>

✓ <https://www.polencapital.com/sites/default/files/Polen-Leveraged-Credit-2025-Review-2026-Outlook.pdf>

✓ <https://www.sikich.com/insight/q4-2025-credit-market-update-year-end-review-and-2026-outlook/>

✓ <https://www.thebanker.com/content/662c67b5-2cdc-4674-a818-1d3692fe8cd7>

✓ <https://www.twentyfouram.com/insights/clos-get-real-on-risk-as-performance-dispersion-rises>

✓ https://www.uscc.gov/sites/default/files/2025-11/Chapter_1--U.S.-China_Economics_and_Trade_Relations_Year_in_Review.pdf

PASS B — ISSUER NEWS 19 / 19 holdings covered (100%)

✓ Ultimate Software Group. (3)

✓ X Corp. (3)

✓ athenahealth Group, In. (3)

✓ Sedgwick Claims Manage. (3)

✓ Mozart Debt Merger Sub. (3)

✓ Peraton Corp. (3)

✓ AI Aqua Merger Sub, In. (3)

✓ McAfee LLC (3)

✓ Proofpoint, Inc. (3)

✓ Bass Pro Group LLC (3)

✓ AmWINS Group, Inc. (3)

✓ Belron Group S.A. (3)

✓ Boxer Parent Co., Inc. (3)

✓ IRB Holding Corp. (3)

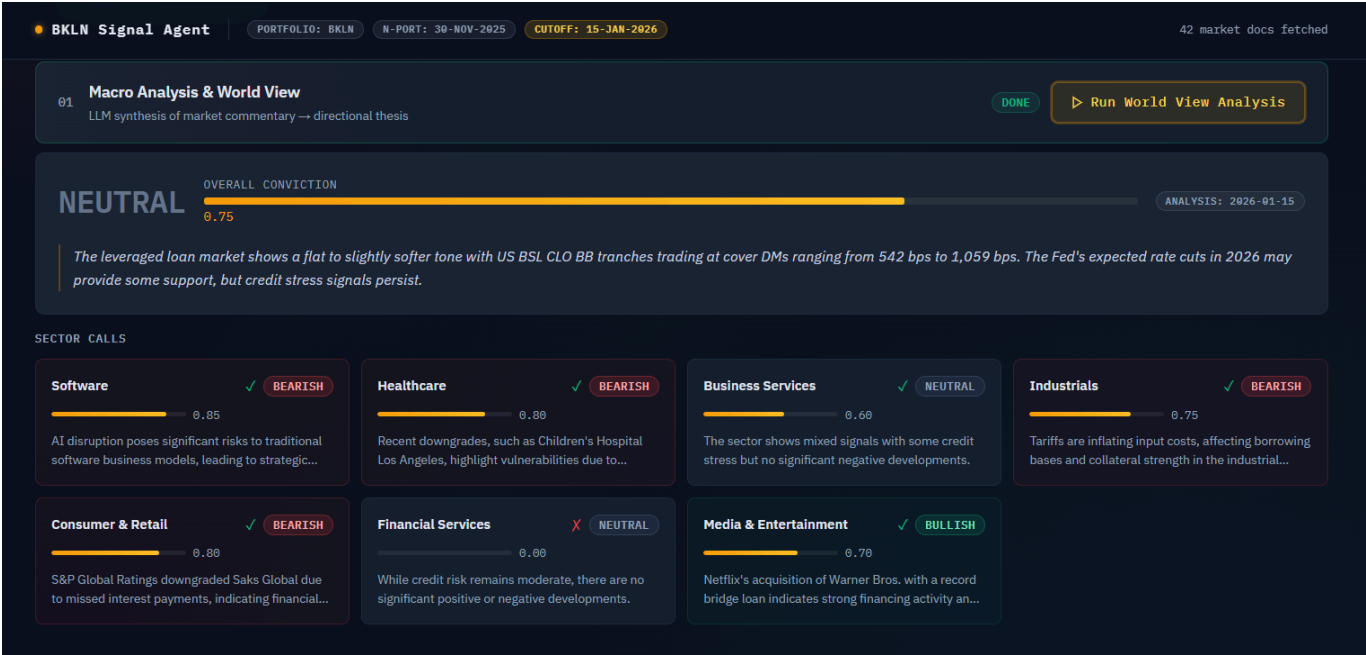
✓ Gainwell Acquisition Co. (3)

✓ Verscend Holding Corp. (3)

✓ Flutter Financing B.V. (3)

✓ Genesys Cloud Services. (3)

✓ Citadel Securities L.P. (3)



BKLN Signal Agent

PORTFOLIO: BKLN

N-PORT: 30-NOV-2025

CUTOFF: 15-JAN-2026

42 market docs fetched

02

Portfolio Impact Mapping

Map thesis onto top-20 holdings → signal + impact scoring

DONE

🗺 Map to Portfolio

SECTOR THESIS (WORLD VIEW)

Software

BEARISH

0.85

AI disruption poses significant risks to tradition...

Healthcare

BEARISH

0.80

Recent downgrades, such as Children's Hospital Los...

Business Services

NEUTRAL

0.60

The sector shows mixed signals with some credit...

Industrials

BEARISH

0.75

Tariffs are inflating input costs, affecting borrowing...

Consumer & Retail

BEARISH

0.80

S&P Global Ratings downgraded Saks Global d...

Financial Services

NEUTRAL

0.80

While credit risk remains moderate, there are no...

Media & Entertainment

BULLISH

0.70

Netflix's acquisition of Warner Bros. with a record...

▲ HIGH CONCENTRATION: Software

38.4% of portfolio

THESIS: BEARISH 0.85

9 holdings

#	ISSUER	SECTOR	WEIGHT	DIRECTION	CONVICTION	SEVERITY	IMPACT (BPS)	EVIDENCE	ACTION
1	Ultimate Software Group, Inc.	SOFTWARE	5.62%	SHORT	0.85	CRITICAL	-22.46	MIXED	Exit / reduce immediately
2	X Corp.	SOFTWARE	5.54%	SHORT	0.95	CRITICAL	-22.14	CORROBORATED	Exit / reduce immediately
3	athenahealth Group, Inc.	SOFTWARE	4.98%	SHORT	0.95	CRITICAL	-19.61	CORROBORATED	Exit / reduce immediately
4	Mozart Debt Merger Sub Inc. (Med...	HEALTHCARE	4.56%	SHORT	0.90	CRITICAL	-18.24	CORROBORATED	Exit / reduce immediately
5	Boxer Parent Co., Inc.	SOFTWARE	3.61%	SHORT	0.95	CRITICAL	-14.45	CORROBORATED	Exit / reduce immediately
6	IRB Holding Corp.	CONSUMER & RETAIL	3.59%	SHORT	0.90	CRITICAL	-14.36	CORROBORATED	Exit / reduce immediately
7	Gainwell Acquisition Corp.	SOFTWARE	3.47%	SHORT	0.95	CRITICAL	-13.90	CORROBORATED	Exit / reduce immediately
8	McAfee LLC	SOFTWARE	4.21%	SHORT	0.70	MEDIUM	-3.16	CONTRADICTED	Watchlist
9	Proofpoint, Inc.	SOFTWARE	4.13%	SHORT	0.70	MEDIUM	-3.10	CONTRADICTED	Watchlist

BKLN Signal Agent

PORTFOLIO: BKLN

N-PORT: 30-NOV-2025

CUTOFF: 15-JAN-2026

42 market docs fetched

PORTFOLIO WEIGHT RECEIVING NEGATIVE SIGNALS

54.9%

EST. DRAWDOWN AVOIDED (SECTOR PROXY)

~135 bps

POSITIONS RATED CRITICAL

7

Ultimate Software Group, Inc., X Corp., athenahealth Group, Inc....

03 Thesis Evaluation

Validate thesis against Jan–Mar 2026 real outcome

DONE

Run Evaluation

A – ESTIMATED LOSS PREVENTION (SECTOR PROXY)

Proxy Portfolio Outcome (Hold All Positions)

~ -321 bps

→

Proxy Outcome After Applying Signals (13 exits)

~ -125 bps

=

Est. Loss Prevented by Acting on Signals

+ ~196 bps

Jan 19 → Mar 20, 2026 · S&P LSTA LL100 · sector-level proxies

B – SECTOR THESIS SCORECARD					4 / 7 HITS
SECTOR	THESIS	CONV	PROXY OUTCOME	RESULT	
Software	BEARISH	0.85	-760 bps	✓ HIT	
Healthcare	BEARISH	0.80	-120 bps	✓ HIT	
Consumer & Retail	BEARISH	0.80	-180 bps	✓ HIT	
Financial Services	NEUTRAL	0.00	-80 bps	✓ HIT	
Business Services	NEUTRAL	0.60	-150 bps	~ NEAR	
Industrials	BEARISH	0.75	-90 bps	✗ MISS	
Media & Entertainment	BULLISH	0.70	-130 bps	✗ MISS	